Phys 210 Solution

HW 08 8.12, 16, 20, 28, 37, 43, 78, 100, 109

8.12 $J_x = \Delta P_x = mV_{2x} - mV_{1x} = (0.145\,kg)(-165\,m/s)\cos 30° - 590\,m/s) = -15.4\,kg\cdot m/s$

$J_y = \Delta P_y = mV_{2y} - mV_{1y} = (0.145\,kg)(165.0\,m/s)\sin 30° - 0) = 4.71\,kg\,m/s$

$F_{av\,x} = \frac{J_x}{\Delta t} = \frac{-15.4\,kg\cdot m/s}{1.75\times10^{-3}\,s} = -8800\,N$ $\quad F_{av\,y} = \frac{J_y}{\Delta t} = \frac{4.71\,kg\cdot m/s}{1.75\times10^{-3}\,s} = 2690\,N$

The horizontal component is 8800N, to the left, and the vertical component is 2690 upward.

8.16 (a) $P_{1x} = P_{2x}$ gives $(0.400\,kg)(10.0\,m/s) = (0.400\,kg + 70.0\,kg)V_2$ and $V_2 = 0.0568\,m$

b) $P_{1x} = P_{2x}$ gives $(0.400\,kg)(10.0\,m/s) = (0.400\,kg)(-8.00\,m/s) + (70.0\,kg)V_{B2}$ $\quad V_{B2} = 0.103$

8.20 (a) $m_A V_{A1x} + m_B V_{B1x} = m_A V_{A2x} + m_B V_{B2x}$ $\quad 0 = m_A V_{A2x} + m_B V_{B2x}$

$V_{A2x} = -\left(\frac{m_B}{m_A}\right)V_{B2x} = -\left(\frac{3.00\,kg}{1.00\,kg}\right)(+1.20\,m/s) = -3.60\,m/s$

Block A has a final speed of 3.60 m/s and moves off in the opposite direction to B

(b) $K_1 + U_1 + W_{other} = K_2 + U_2$. only the spring force does work so $W_{other} = 0$ and $U = U_e$

$K_1 = 0$ (the blocks initially are at rest)

$U_2 = 0$ (no potential energy is left in the spring)

$K_2 = \frac{1}{2}m_A V_{A2}^2 + \frac{1}{2}m_B V_{B2}^2 = \frac{1}{2}(1.00\,kg)(3.60\,m/s)^2 + \frac{1}{2}(3.00\,kg)(1.20\,m/s)^2 = 8.64\,J$

Thus $U_{i,el} = K_2 = 8.64\,J$

8.28 (a) $P_{1x} = P_{2x}$ gives $mV_{A1} = mV_{A2}\cos 30.0° + mV_{B2}\cos 45.0°$

$P_{1y} = P_{2y}$ gives $0 = mV_{A2}\sin 30.0° - mV_{B2}\sin 45.0°$

Combining these two equations gives $0.500\,V_{A2} = 40.0\,m/s - 0.866\,V_{A2}$

$V_{A2} = 29.3\,m/s$

$V_{B2} = \left(\frac{0.500}{0.707}\right)(29.3\,m/s) = 20.7\,m/s$

(b) $K_1 = \frac{1}{2}mV_{A1}^2$ $\quad K_2 = \frac{1}{2}mV_{A2}^2 + \frac{1}{2}mV_{B2}^2$

$\frac{K_2}{K_1} = \frac{V_{A2}^2 + V_{B2}^2}{V_{A1}^2} = \frac{(29.3\,m/s)^2 + (20.7\,m/s)^2}{(40.0\,m/s)^2} = 0.804$

$\frac{K_2 - K_1}{K_1} = \frac{K_2}{K_1} - 1 = -0.196$

of the original kinetic energy is dissipated during this collision

8.37 $P_{1x} = P_{2x}$ gives $m_A V_{A1x} = (m_A + m_B) V_{2x}$

$$V_{A1x} = \left(\frac{m_A + m_B}{m_A}\right) V_{2x} = \left(\frac{950\,kg + 1900\,kg}{950\,kg}\right)(6.5\,m/s) = 19.5\,m/s$$

$P_{1y} = P_{2y}$ gives $m_B V_{B1y} = (m_A + m_B) V_{2y}$

$$V_{B1y} = \left(\frac{m_A + m_B}{m_B}\right) V_{2y} = \left(\frac{950\,kg + 1900\,kg}{1900\,kg}\right)(4.6\,m/s) = 21.9\,m/s$$

Before the collision the subcompact car has speed 19.5 m/s and the truck has speed 21.

8.43 (a) Conservation of momentum gives

$(0.0300\,kg)(0.200\,m/s) + (0.100\,kg)(-0.400\,m/s) = (0.0300\,kg)V_{A2x} + (0.0100\,kg)V_{B2x}$

$3V_{A2x} + V_{B2x} = 0.200\,m/s$

Eq. 8.27 says $V_{B2x} - V_{A2x} = -(-0.400\,m/s - 0.200\,m/s) = 0.600\,m/s$

Solving this pair of equations gives $V_{A2x} = -0.100\,m/s$ $V_{B2x} = 0.500\,m/s$

The 30.0 g marble is moving to the left at 0.100 m/s and the 10.0 g marble is moving to the right at 0.500 m/s.

(b) For marble A, $\Delta P_{Ax} = m_A V_{A2x} - m_A V_{A1x} = (0.0300\,kg)(-0.100\,m/s - 0.200\,m/s) = -0.00900\,kg\,m$

For marble B, $\Delta P_{Bx} = m_B V_{B2x} - m_B V_{B1x} = (0.0100\,kg)(0.500\,m/s - (-0.400\,m/s)) = +0.00900\,kg\,m$

The changes in momentum have the same magnitude and opposite sign.

(c) For marble A, $\Delta K_A = \frac{1}{2} m_A V_{A2}^2 - \frac{1}{2} m_A V_{A1}^2 = \frac{1}{2}(0.0300\,kg)[(0.100\,m/s)^2 - (0.200\,m/s)^2] = -4.5 \times 10^{-4}\,J$

For marble B, $\Delta K_B = \frac{1}{2} m_B V_{B2}^2 - \frac{1}{2} m_B V_{B1}^2 = \frac{1}{2}(0.0100\,kg)[(0.500\,m/s)^2 - (0.400\,m/s)^2] = +4.5 \times 10^{-4}\,J$

The changes in kinetic energy have the same magnitude and opposite sign.

8.78 Conservation of energy says $\frac{1}{2} m v^2 = mgR$, $v = \sqrt{2gR}$

Conservation of momentum applied to the collision gives $mv_1 = 2mv_2$, $v_2 = \sqrt{\frac{gR}{2}}$

$\frac{1}{2}(2m)v_2^2 = (2m)g y_3$

$$y_3 = \frac{v_2^2}{2g} = \frac{1}{2g}\frac{gR}{2} = \frac{R}{4}$$

8.100 Initially $v_{cm1} = 0$, so the center of mass of the canoe doesn't move.

Initially $x_{cm1} = \frac{m_A x_{A1} + m_B x_{B1}}{m_A + m_B}$ After she walks $x_{cm2} = \frac{m_A x_{A2} + m_B x_{B2}}{m_A + m_B}$

$x_{cm1} = x_{cm2}$ gives $m_A x_{A1} + m_B x_{B1} = m_A x_{A2} + m_B x_{B2}$

She walks to a point 1.00 m from the right-hand end of the canoe, so she is 1.50 m to the right of the center of mass of the canoe and $x_{A2} = x_{B2} + 1.50\,m$

$(45.0\,kg)(0.00\,m) + (60.0\,kg)(3.00\,m) = (45.0\,kg)(x_{B2} + 1.50\,m) + (60.0\,kg)x$

$105.0\,kg)\ x_{B2} = 127.5\,kg\cdot m$ and $x_{B2} = 1.21\,m$

$x_{B2} - x_{B1} = 1.21\,m - 2.50\,m = -1.29\,m$ The canoe moves $1.29\,m$ to the left.

8.109 a) $U_1 = K_2$ says $m_A g\,(50\,m)(\sin 6.0°) = \frac{1}{2} m_A v_{A1}^2$

$v_{A1} = 10.12\,m/s$ is the speed of the wagon just before the collision

$P_{1x} = P_{2x}$ for the collision says $m_A v_{A1} = (m_A + m_B) v$

$$v = \left(\frac{300\,kg}{300\,kg + 75.0\,kg + 60.0\,kg}\right)(10.12\,m/s) = 6.98\,m/s$$

In 5.0 s the wagon travels $(6.98\,m/s)(5.0\,s) = 34.9\,m$, and the people will ha time to jump out of the wagon before it reaches the edge of the cliff.

b) For the wagon, $K_1 = \frac{1}{2}(300\,kg)(10.12\,m/s)^2 = 1.54 \times 10^4\,J$

Assume that the motorcycles drop from a small height so their kinetic energy ju before the wagon can be neglected compared to K_1 of the wagon.

$K_2 = \frac{1}{2}(435\,kg)(6.98\,m/s)^2 = 1.06 \times 10^4\,J$

The kinetic energy of the system decreases by $K_1 - K_2 = 4.8 \times 10^3\,J$